MUSFIQUE US SALEHIN



Computer Science and Engineering student with a solid foundation in core programming, software design, and algorithmic problem-solving. Quick to learn and apply new technologies across full-stack development, software Engineering and testing, data systems, and technical operations. Seeking an entry-level technical role where adaptable technical skills and clear communication add immediate value.

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Education

BRAC University | Bachelor of Science in Computer Science & Engineering | *Expected Dec 2026*

Govt. Hazi Muhammad Mohsin College | HSC, Chattogram | *GPA: 5.00 / 5.00* | *2020*

Ispahani Public School & College | SSC, Chattogram | *GPA: 5.00 / 5.00* | *2018*

Experience

Rider Service Agent, Operations | *Foodpanda* | *Dhaka, Bangladesh* | *December 2024 – Present*

- Onboarded 8,000+ riders through outbound recruitment calls, averaging 100 calls/day across hiring, reactivation, and fraud follow-up.
- Resolved 10,000+ rider issues, working with the customer experience team to keep service consistent across the platform.
- Identified and addressed fraudulent rider and customer activity by analyzing call and platform data, protecting platform integrity.
- Maintained accurate operational records and consistently met daily team targets in a high-volume environment.

Projects

PetOn | *Full-Stack Pet Session Booking Platform* | Live Demo | GitHub Repository | *2026*

React.js, Vite, Node.js, Express, MongoDB, ImageKit, JWT, Tailwind CSS

- Developed a dual-portal management system featuring a client-side interface for pet session discovery and an administrative dashboard for listing pets.
- Engineered a dynamic search and filtering engine, allowing users to query pets by species, availability, and session type.
- Implemented Role-Based Access Control (RBAC) to secure administrative actions.
- Integrated a real-time booking workflow with state management to prevent double-booking.

VisDrone-Object-Detection | *AI-Powered Aerial Detection & Tracking Pipeline* | GitHub Repository | *2026*

Python, PyTorch, YOLOv8, ByteTrack, Supervision, OpenCV, Cloud GPU (T4)

- Trained and fine-tuned a custom **YOLOv8** object detection architecture on the high-altitude VisDrone dataset, consolidating 10 original visual classes into refined human and vehicle targets.
- Integrated **ByteTrack** multi-object tracking via the Supervision library, executing persistent frame-by-frame data association to sustain tracking IDs.
- Engineered a real-time tracking metrics pipeline with a customized Heads-Up Display (HUD), using Python set structures to filter frames and output distinct moving-target totals without duplicate counting.

Dhaka Tesla Pool | *Concurrency-Safe Ride-Pooling Platform* | Live Demo | GitHub Repository | *2026*

React.js, Vite, Node.js, Express.js, PostgreSQL, Knex, Docker, JWT, Tailwind CSS

- Engineered a ride-pooling backend where concurrent requests for a vehicle's last open seat are resolved safely using row-level locking and database CHECK constraints, verified by an automated test that races two simultaneous requests across 8 runs against PostgreSQL.
- Modeled the ride and vehicle-pool lifecycle as two independent state machines to support per-passenger drop-off and fare-locking within one shared trip, with every transition validated server-side and rejected with structured errors when invalid.
- Diagnosed and resolved a transaction deadlock caused by inconsistent lock ordering across driver commands, restoring a single consistent vehicle-to-pool-to-ride locking order throughout the service layer.
- Containerized the full stack with a multi-stage Docker build and deployed it to Render with a managed PostgreSQL instance (Neon), backed by a 92-test suite covering authentication, fare calculation, and booking integrity.

Research & Technical Achievements

- **Sickle Cell Detection (Undergraduate Thesis):** Developed a highly efficient diagnostic pipeline (1.4M parameters) using a MobileNetV3-Small encoder and Temporal-Difference Multiple Instance Learning (TD-MIL) to classify sickle cell disease at the donor level. Achieved a 0.990 macro-AUC and 100% recall with strict donor-level cross-validation, optimized for deployment on low-cost clinical hardware.
- **Geo-Restore GAN:** Developing a dual-head adversarial framework that restores image quality in adverse weather while preserving metric depth accuracy, aimed at autonomous-vehicle perception safety.
- **Problem Solving:** Solving problems across competitive programming platforms in Python and C++, focusing on data structures, algorithms, and Object-Oriented Programming (OOP).

Skills

- **Languages:** Python, JavaScript, C++, HTML, CSS
- **Frontend, Backend & Databases:** React.js, Tailwind CSS, Node.js, Express.js, MongoDB, MySQL, PostgreSQL
- **AI / Machine Learning:** TensorFlow, PyTorch, OpenCV, YOLOv8
- **Tools & Platforms:** Git, GitHub, Docker, Linux, Postman, Figma, Canva, MS Office
- **Soft Skills & Languages:** Communication, Presentation, Public Speaking | English, Bengali

Extracurricular Activities, Sports & Awards

- **Secretary of Finance & Event Mgt.** | *BRAC University Arts & Photography Society*
  Managed budget allocations and operational logistics for university-wide photography exhibitions and cultural festivals.
- **Junior Executive of Arts & Design** | *Robotics Club of BRAC University*
  Designed visual assets, event branding, and promotional banners for robotics competitions and technical workshops.
- **Sports & Athletics:** Won 15+ school athletics medals; 4-time group champion and 2× Best Athlete award recipient.